

Linear equations and checking systems

While TSM 099 studies Topic 32. Review: Topic 24, Topic 25, Topic 26, Topic 28, Topic 31.

Equivalent forms are acceptable. Use these answers after attempting the problems; revisit the linked lecture notes for additional examples.

Core: two-term and short linear equations

1. Solve and check in the original equation.

$$-6x - 2 = 16$$

$$x = -3$$

Check by substituting into the original equation.

2. Solve and check in the original equation.

$$-3x - 1 = -7$$

$$x = 2$$

Check by substituting into the original equation.

3. Solve and check in the original equation.

$$-6x - 7 = 17$$

$$x = -4$$

Check by substituting into the original equation.

4. Solve and check in the original equation.

$$-3x + 8 = 8$$

$$x = 0$$

Check by substituting into the original equation.

5. Solve and check in the original equation.

$$-6x + 8 = -28$$

$$x = 6$$

Check by substituting into the original equation.

6. Solve and check in the original equation.

$$2x + 3 = 11$$

$$x = 4$$

Check by substituting into the original equation.

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Core: variable terms on both sides

7. Solve and check in the original equation.

$$-5x + 1 = x - 8$$

$$x = \frac{3}{2}$$

Check by substituting into the original equation.

8. Solve and check in the original equation.

$$4x + 6 = x + 1$$

$$x = -\frac{5}{3}$$

Check by substituting into the original equation.

9. Solve and check in the original equation.

$$-3x - 6 = 5x + 2$$

$$x = -1$$

Check by substituting into the original equation.

10. Solve and check in the original equation.

$$-5x + 1 = -2x - 5$$

$$x = 2$$

Check by substituting into the original equation.

11. Solve and check in the original equation.

$$-3x - 7 = 5x + 5$$

$$x = -\frac{3}{2}$$

Check by substituting into the original equation.

12. Solve and check in the original equation.

$$-5x + 1 = -2x + 3$$

$$x = -\frac{2}{3}$$

Check by substituting into the original equation.

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Core: review linear equations with fractions

13. Solve and check in the original equation.

$$\frac{5x - 14}{6} = \frac{-2x - 5}{6}$$

$$6(5x - 14) = 6(-2x - 5)$$

$$x = \frac{9}{7}$$

Check by substituting into the original equation.

14. Solve and check in the original equation.

$$\frac{5x - 8}{3} = -1$$

$$1(5x - 8) = 3(-1)$$

$$x = 1$$

Check by substituting into the original equation.

15. Solve and check in the original equation.

$$\frac{-13x + 13}{10} = \frac{-x + 1}{8}$$

$$8(-13x + 13) = 10(-x + 1)$$

$$x = 1$$

Check by substituting into the original equation.

16. Solve and check in the original equation.

$$\frac{3x - 4}{12} = 0$$

$$1(3x - 4) = 12(0)$$

$$x = \frac{4}{3}$$

Check by substituting into the original equation.

17. Solve and check in the original equation.

$$\frac{x - 11}{8} = \frac{-2x - 6}{5}$$

$$5(x - 11) = 8(-2x - 6)$$

$$x = \frac{1}{3}$$

Check by substituting into the original equation.

18. Solve and check in the original equation.

$$\frac{6x + 1}{6} = \frac{-2x - 3}{2}$$

$$2(6x + 1) = 6(-2x - 3)$$

$$x = -\frac{5}{6}$$

Check by substituting into the original equation.

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Extension: check proposed solutions of systems

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- 19.** Does the ordered pair solve both equations?
Show both substitutions.

$$1(2) + (1)(3) = 5; \quad 2(2) + (-1)(3) = 1$$

Yes; both equations are true.

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- 20.** Does the ordered pair solve both equations?
Show both substitutions.

$$1(3) + (-1)(-1) = 4; \quad 2(3) + (1)(-1) = 5$$

No; at least one equation is false.

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- 21.** Does the ordered pair solve both equations?
Show both substitutions.

$$2(2) + (3)(-1) = 1; \quad 1(2) + (-2)(-1) = 4$$

Yes; both equations are true.

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- 22.** Does the ordered pair solve both equations?
Show both substitutions.

$$1(1) + (2)(3) = 7; \quad 3(1) + (-1)(3) = 0$$

Yes; both equations are true.

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- 23.** Does the ordered pair solve both equations?
Show both substitutions.

$$2(3) + (-1)(1) = 5; \quad 1(3) + (1)(1) = 4$$

Yes; both equations are true.

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- 24.** Does the ordered pair solve both equations?
Show both substitutions.

$$3(-1) + (1)(1) = -2; \quad 1(-1) + (-1)(1) = -2$$

Yes; both equations are true.